Assignment No 2

Aim: Develop a Java program for an E-commerce order processing where some products are initialized through multiple constructors, users can input some product details manually, the system computes total order cost dynamically, applies discount policies based on conditions, and presents a detailed invoice summarizing the purchase

Code:

```java
import java.util.Scanner;

class Product {
    int productId;
    String productName;
    double price;
    int quantity;

    Product() {
        productId = 0;
        productName = "Unknown";
        price = 0;
        quantity = 0;
    }

    Product(int productId, String productName, double price) {
        this.productId = productId;
        this.productName = productName;
        this.price = price;
        this.quantity = 1;
    }

    Product(int productId, String productName, double price, int quantity) {
        this.productId = productId;
        this.productName = productName;
        this.price = price;
        this.quantity = quantity;
    }

    double getTotalPrice() {
        return price * quantity;
    }

    double getDiscount() {

        if (getTotalPrice() <= 0) {
            return 0.0;
        }
        if (getTotalPrice() > 50000) {
            return 0.20;
        } else if (getTotalPrice() > 25000) {
            return 0.10;
        }else {
            return 0.05;
        }
    }
```

```java
    void displayProduct() {
        System.out.println("\n----------- Product Details ----------");
        System.out.println("Product ID\tName\t\tQty\tPrice\tTotal");
        System.out.println("---------------------------------------");
        System.out.println("Product Id : " + productId);
        System.out.println("Product Name : " + productName);
        System.out.println("Product Price : " + price);
        System.out.println("Product quantity : " + quantity);
        System.out.println("Total Amount : ₹" + getTotalPrice());
        System.out.println("Discount     : ₹ " + getDiscount());
        System.out.println("Final Amount : ₹" + (getTotalPrice() - getTotalPrice() * getDiscount()));
        System.out.println("----------------------------------------");
    }
}

public class Ecommerce {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        Product p1 = new Product();
        Product p2 = new Product(102, "Mouse", 45000);
        Product p3 = new Product(103, "Keyboard", 500, 2);

        p1.displayProduct();
        p2.displayProduct();
        p3.displayProduct();

        System.out.print("Enter Product ID: ");
        int id = sc.nextInt();

        System.out.print("Enter Product Name: ");
        String name = sc.next();

        System.out.print("Enter Price: ");
        double price = sc.nextDouble();

        System.out.print("Enter Quantity: ");
        int qty = sc.nextInt();

        Product userProduct = new Product(id, name, price, qty);
        userProduct.displayProduct();
    }
}
```

Output:

```
----------- Product Details ----------
Product ID  Name        Qty Price   Total
----------------------------------------
Product Id : 0
Product Name : Unknown
Product Price : 0.0
Product quantity : 0
Total Amount : ₹0.0
Discount     : ₹ 0.0
Final Amount : ₹0.0
-----------------------------------------

----------- Product Details ----------
Product ID  Name        Qty Price   Total
----------------------------------------
Product Id : 102
Product Name : Mouse
Product Price : 45000.0
Product quantity : 1
Total Amount : ₹45000.0
Discount     : ₹ 0.1
Final Amount : ₹40500.0
-----------------------------------------
```

```
----------- Product Details ----------
Product ID  Name        Qty Price   Total
----------------------------------------
Product Id : 103
Product Name : Keyboard
Product Price : 500.0
Product quantity : 2
Total Amount : ₹1000.0
Discount     : ₹ 0.05
Final Amount : ₹950.0
-----------------------------------------
Enter Product ID: 105
Enter Product Name: Sony
Enter Price: 85000
Enter Quantity: 2

----------- Product Details ----------
Product ID  Name        Qty Price   Total
----------------------------------------
Product Id : 105
Product Name : Sony
Product Price : 85000.0
Product quantity : 2
Total Amount : ₹170000.0
Discount     : ₹ 0.2
Final Amount : ₹136000.0
-----------------------------------------
```